



Nathan Ivor Sequeira

Date of birth: 02/12/2003 | **Nationality:** Indian | **Gender:** Male | **Email address:** nathansequeirade@gmail.com |
Website: <https://nathansequeirafinal.vercel.app/> | **Address:** India (Home)

● ABOUT ME

BCA graduate with 1.5 years of professional experience in DNS, VPS, and VPN infrastructure support, including server-side monitoring, network fault diagnosis, and QA testing. Independent project work spans distributed systems, ML-based anomaly detection, and event-driven backend architecture. Seeking to formalise and deepen that foundation through the M.Eng. AI and Advanced Information Technologies programme at Hochschule RheinMain.

● EDUCATION AND TRAINING

19/10/2021 – 22/05/2024 Mangaluru, India

BACHELOR OF COMPUTER APPLICATION St. Aloysius College (Autonomous)

Studied core areas including Data Structures and Algorithms, Object-Oriented Programming (Java, C, C++), Database Systems, Operating Systems, Computer Networks, AI and Machine Learning, Software Engineering, and Mathematics including Linear Algebra, Probability, and Statistics. Final CGPA: 6.84/10 (German equivalent: 2.5), with Semester VI returning an SGPA of 8.10, the strongest semester of the degree.

Final grade CGPA: 6.84/10 — German equivalent: 2.5 | **Level in EQF** EQF level 6

● WORK EXPERIENCE

JUNIOR SUPPORT ENGINEER - BLUEHOST - UNIFYCX – 04/07/2024 – 19/03/2025 – MANGALURU, INDIA

- Diagnosed and resolved DNS propagation failures, SSL certificate errors, and email delivery breakdowns across shared and VPS hosting environments using WHM, cPanel, and Linux-based tools.
- Used Grafana dashboards daily to correlate server-side symptoms with client-reported failures, identifying root causes across hosting, DNS, and security layers.

JUNIOR ASSOCIATE - OPERATIONS - TECHNICAL SUPPORT - UNIFYCX – 20/03/2025 – 04/11/2025 – MANGALURU, INDIA

- Served as designated QA tester for the team: designed test cases, documented failure modes, and validated fixes across VPN infrastructure and other products for PointWild, a VPN company, following release to identify and resolve post-launch issues.
- Resolved 40-80 VPN support tickets per shift, diagnosing authentication failures, tunnel configuration errors, and server load imbalances using Zendesk and Okta.

● SKILLS

Programming Languages

SQL | JavaScript | Python | C and C++

Infrastructure

cPanel | Docker | VPS | WHM | DNS | Linux

ML & Data

Isolation Forest | NVIDIA NIM | Google Gemini Pro | scikit-learn | OpenAI API | anomaly detection pipelines

Frameworks

ReactJS | Next.js | LangChain | Tailwind CSS | FastAPI

Databases

SQLite | PostgreSQL | MySQL | Pinecone

DevOps / CI

Git | GitHub | GitHub Actions

Observability

Prometheus | Grafana

Security

HMAC-SHA256 | Palo Alto Networks

● PROJECTS

InfraBoard — Cloud-Native Microservices Observability Dashboard (FastAPI, Prometheus, Grafana, Docker)

- Built a three-microservice observability system where Prometheus scrapes metrics every 15 seconds, Grafana visualises them, and an unsupervised Isolation Forest model detects anomalies and persists alerts to SQLite. The choice of unsupervised detection was deliberate as no labelled anomaly dataset existed, so the model learns normal behaviour from recent metric history.
- Documented the Docker Compose over Kubernetes decision in an ADR, with a clear migration path for when scale requires it.
- GitHub: <https://github.com/Nathan-sudo-pycharm/infraboard-observability>

CodeReviewBot — AI-Powered GitHub PR Reviewer (FastAPI, Celery, Redis, Mistral, Next.js)

- Event-driven async system where GitHub webhook events trigger a Celery and Redis task queue, diff analysis runs via Mistral through NVIDIA NIM, and severity-ranked review comments are posted directly to Pull Requests.
- Implemented HMAC-SHA256 webhook signature verification to prevent event spoofing from external actors.
- GitHub Link: <https://github.com/Nathan-sudo-pycharm/codereviewbot>

Hermes — Distributed Workflow Orchestration Platform (Python, Message Queue, PostgreSQL)

- Built a self-hosted distributed workflow orchestration platform where tasks are queued through Apache Kafka, distributed to workers, and results reported back to the coordinator via gRPC. The system implements a per-worker circuit breaker state machine (CLOSED, OPEN, HALF-OPEN), retry with exponential backoff, a dead-letter queue for exhausted tasks, and OpenTelemetry distributed tracing propagated across Kafka and gRPC boundaries into Jaeger.
- Load tested under concurrent users with zero failure rate. All major architectural decisions documented in ADRs covering Kafka over Redis Streams, gRPC over REST for internal communication, and circuit breaker placement in the coordinator.
- GitHub Link: <https://github.com/Nathan-sudo-pycharm/hermes>

● CERTIFICATIONS

Honeywell & Palo Alto Networks, 31/10/2023

Honeywell Cybersecurity Training (Grade A)

Completed certification covering cybersecurity fundamentals, firewall operations, and cyber attack prevention strategies. Training delivered by Honeywell and Palo Alto Networks at St. Aloysius College.

Palo Alto Networks, 31/10/2023

Cloud Security Fundamentals — Palo Alto Networks

Completed foundational training in cloud security principles, covering threat detection, network security, and cloud infrastructure protection.